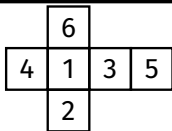
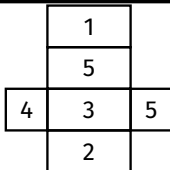


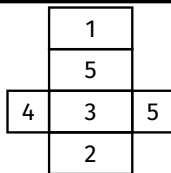
# Push Tasks - Nets



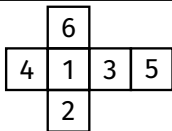
- 1. Invalid arrangements?**  
e.g. 4 in a row with 2 on same side — overlaps



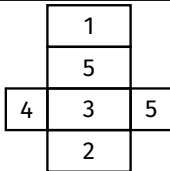
- 3. 3 pairs face areas?**  
15, 10, 6 each x 2



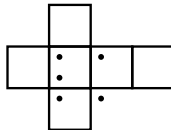
- 5. Max SA for V=96?**  
 $1 \times 1 \times 96 = 386 \text{ cm}^2$



- 2. Why some fail?**  
Adjacent squares overlap

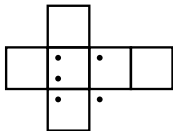


- 4. Min SA for V=96?**



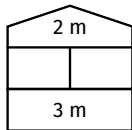
- 6. Opposite sums to 7?**  
Standard: 1-6 2-5 3-4

# Push Tasks - Nets



**7. Different die net?**

**Any valid cube net works**



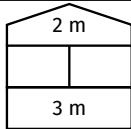
**9. Different net?**

**Swap triangle positions**



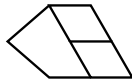
**11. Rectangular pyramid?**

**Net: 1 rect + 4 triangles (2 pairs)**



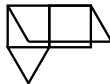
**8. Triangles isosceles?**

**Two equal sides by**



**10. Triangles congruent?**

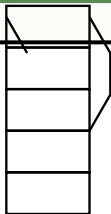
**Yes — same base and slant height**



**12. Right triangle prism?**

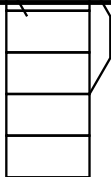
**Rectangles + 2 right triangles**

# Push Tasks - Nets



13. Distinct nets?

Many — 2 hexagons + 6



14. Hex prism SA?



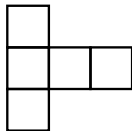
15. Tetrahedron nets?

2 distinct nets



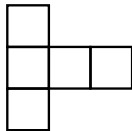
16. Why 4 triangles?

4 faces, all triangles



17. Lid positions?

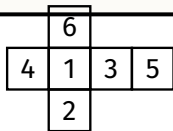
Depends on shape



18.  $V=12$  no top SA?

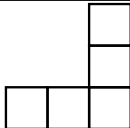
Depends on dimensions

# Push Tasks - Nets



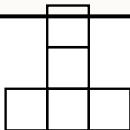
**19. Shortest surface path?**

**Unfold net, draw straight line**



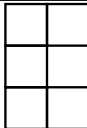
**20. L as cube net?**

**Only if 6th square added**



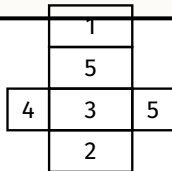
**21. 2 non-box pentominoes?**

**I-pentomino (straight) and X-pentomino**



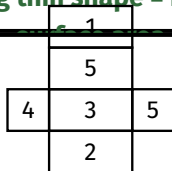
**22. All open box nets?**

**11 pentominoes can form**



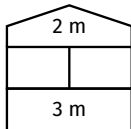
**23. 1x1x96 max paper?**

**Long thin shape = huge**



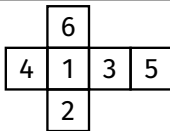
**24. 4x6x4 min paper proof?**

# Push Tasks - Nets



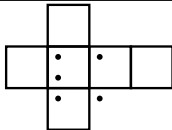
**25. Ribbon on faces?**

**3.8 m on faces, 0.2 m knot**



**27. Shortest path?**

**Draw net, connect start  
to end**



**26. Min colours?**

**3 (opposite faces same)**